

Vibra-Volt Pedal Cover Set

Cost Analysis per Unit — CSMFAB 026

Category: Aegis Embark — Pedals **Unit Basis:** 1 set (accelerator + brake, fits standard EV) **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
ZrB2-SiC ceramic tip inserts (2pc)	0.15 kg	\$350.00/kg	\$52.50
KNbO3-BaTiO3 harvesting stack (2 sets, 32 elements)	0.03 kg	\$180.00/kg	\$5.40
BFRP snap-fit base shells (2pc, Elium®)	0.3 kg	\$25.00/kg	\$7.50
FBG optical sensor elements (4×)	0.01 kg	\$120.00/kg	\$1.20
PMMA plastic optical fiber (1m)	0.05 kg	\$25.00/kg	\$1.25
Supercapacitor buffer (10F, 5V)	0.05 kg	\$35.00/kg	\$1.75
YInMn Blue glaze (ceramic tip)	0.02 kg	\$75.00/kg	\$1.50
Universal mounting clip hardware (per size)	0.06 kg	\$45.00/kg	\$2.70
BOM Subtotal			\$73.80

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$73.80	\$60.52	\$47.97
Direct Labor	\$85.00	\$58.00	\$38.00
Overhead (15% of labor)	\$12.75	\$8.70	\$5.70
Quality Control & Testing	\$10.20	\$4.64	\$1.90
Packaging & Logistics	\$25.00	\$20.00	\$15.00

Tooling & Equipment Notes: Ceramic tip mold: \$8,000 NRE. FBG sensor is off-shelf telecom component.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$580	\$928 – \$1,276	35-55%
Small Batch (100-500/yr)	\$390	\$585 – \$780	33-50%
Production Volume (5K-50K/yr)	\$275	\$385 – \$495	28-45%

Market Reference: \$320-480 (EV accessories market; first GIC-immune pedal cover — unopposed market entry)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$13 – \$25 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction